PROJECT-BASED LEARNING

PHASE-I EVALUATION

The Detective's Dossier: An Interactive 2D Murder Mystery and Investigation Game

Team ID: DSCPP-III-2026-T027

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T027
Semester: 3rd
Section: ML1, ML2, ML3
Domain: *DSA/OOPs with C++*

Team Members

1. *Ashutosh Bahuguna – 2510370440*
2. *Snehal Pundir – 2510370590*
3. *Vidhi Somraan – 2510380302*

Mentor

Mr. Sujal Bindra

Interactions completed: 03

Problem Statement & Motivation

Problem Statement

Our project enables the players to connect with the clues , suspects, the evidences, and the locations, and interact with them to make it a more interactive game.

Motivation

- *limited opportunities to independently connect clues, suspects, and locations*
- *simplified investigation reduces logical reasoning and, evidence analysis.*
- *predefined narratives limit player-driven investigation.*

Target Users / Stakeholders

- *students who enjoy mystery and detective games*
- *gamers who like logical, thinking and problem solving*
- *players seeking an interactive mystery-solving experience*

Objectives

Project Objectives

1. *To develop 2D interactive investigation game where the players investigate and solve the crime cases .*
2. *Apply the C++ concepts using OOPs to organize the cases, evidences , locations and game modules .*
3. *Implement the DSA concepts such as graphs,DFS/BFS , Dijkstra ,hashing, stacks,queues, searching and sorting.*
4. *Evaluate the system using suitable test cases.*
5. *Provide a case-resolution mechanism where players connect evidence and make a final accusation.*

Key Objective: *enabling interactive evidence-based problem solving*

Proposed Solution

Proposed Approach

1. *player actions, clues, suspects and locations*
2. *process clues, investigate relations*
3. *manage cases, puzzles, suspects,evidences*
4. *store clues, suspects, locations, case data*
5. *display investigation results*

Key Features

- *interactive 2D investigation interface*
- *clue evidence management*
- *suspect–clue relations analysis*
- *puzzle-based investigation*
- *case progress final result*

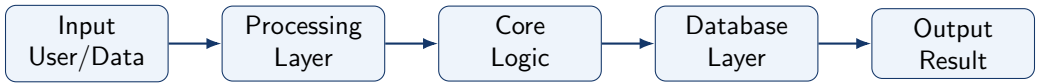
Clearly explain how the proposed solution addresses the identified problem.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	*Stacks, Queue,Graphs, Hashing, Searching*	*manage, organize the game data*
OOPs with C++	*Classes, Inheritance, Polymorphism*	*code organize*

Requirement: meaningful integration of both subjects DSA and OOPs with C++ concepts in **one** project.

System Architecture / Workflow



Major Components

- *2D game, user interface database and case management*
- *investigation,DSA mechanism*
- *Sqlite database,case management*

Data / Control Flow

- *playerinputs→Qtinterface→ C++/OOP gameplay→DSA investigation→SQLite database→result*
- *game interface communicates with the core logic, which uses DSA and database modules*
- *processed clues, identified suspect,final case outcome*

Technology Stack & Methodology

Technology Stack

- Programming: *DSA/C++/OOPs*
- Database: *SQLite*
- Tools / Framework: *Qt*
- IDE: *Qt creator*
- Version Control: *Git/GitHub*

Development Methodology

1. Requirement Analysis
2. System Design
3. Implementation
4. Testing
5. Evaluation

Justify the selection of technologies and methodology.

Initial Progress & Team Contribution

Progress Achieved

- *studied existing murder mystery,detective games*
- *identified required features, gameplay/investigation.*
- *designed the system architecture workflow.*
- *started designing the C++/OOP game and Qt-based interface.*
- *SQLite database structure for cases,clues, suspects,locations*

Individual Contribution

Member	Role	Contribution
Ashutosh Bahuguna	*Lead*	*Requirements,system architecture, DSA identification*
Snehal Pundir	*Developer*	*Game workflow, UI/gameplay planning*
Vidhi Somraan	*DB/Testing*	*C++/OOP design,SQLite,testing strategy*

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I:project planning, technology selection
2. Phase-II:Qt-based 2D interface, C++/OOP framework,DSA,SQLite database,gameplay modules
3. Phase-III:combining modules,creation,testing investigation cases

Expected Outcomes

- *functional interactive detective game.*
- *engaging platform for logical reasoning, problem-solving*
- *support for new cases, clues, locations,gameplay*

References

1. *R. Lafore, Object-Oriented Programming in C++*
2. *class notes,refernce books*
3. *Qtdocumentation: https://doc.qt.io/qtcreator/*
4. *our GITHub repository contain source code,dataset*

Thank You
Questions & Suggestions